Practice session for Econometrics for Public Policy

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Introduction

Introduction

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Course Information

- This course is a companion course of Econometrics for Public Policy by Professor Chen
  - Delivery type: Hybrid
  - Course time: Wed, period 3 (13:00 - 14:45)
- Zoom meeting information:
  - Meeting ID: 353 650 7388
- GraSPP students who register for Econometrics for Public Policy **are required** to take this course

Course Information

- The goal of this course is to teach you how to use R to conduct empirical research
- Basically, we need **empirical toolkits**, **data**, and **econometrics software** to conduct empirical research
- In this course, you will learn how to do data cleaning work and conduct applied economic research using R
  - Basic commands
  - dplyr, tidyr, ggplot2
  - Advanced packages for causal analysis
  - estimatr, rdd, fixest, didimputation, did
- You can find the course materials in this google drive
- Please use your ECCS Cloud Mail to get access

Course Logistics

- Professor Chen will teach the theoretical part of this course, so I will focus on the **R implementation**
- Basically, this course is divided into two parts:
  - Basic techniques for cleaning, analyzing, and plotting data
  - Empirical commands for causal inferences

Homework and Grading

- There are six homeworks this semester
  1. HW1 (4/16 - 5/7)
  2. HW2 (5/7 - 5/21)
  3. HW3 (5/21 - 6/4)
  4. HW4 (6/4 - 6/18)
  5. HW5 (6/18 - 7/2)
  6. HW6 (7/2 - 7/15)
- Please submit your homework to the grader by email
  - Grader: Jovey Mae DOMINGO
  - Email: grasppmetrics2025@gmail.com
  - If you have any problems with the homework, please contact the grader using the email above

Homework and Grading

- Please submit both the R script or Markdown file and compiled report or knit file
- Late submissions will be discounted by 50% per day
- You can discuss homework with your classmates, but each individual must submit
- chatGPT: use it as you see fit; however, the code submitted must reflect the lectures
- **You won't get credit just because your code runs the correct output!**
- Course Evaluation:
  - HW1 through HW5 = 15% each
  - HW6 = 25%

Outline for Today

- Introduction to R
- Access to R
- Basics of R
  - Common functions
  - Interface
  - Type commands
  - Typical format
  - Some basic commands
- Homework 0

Introduction to R

Installation and setup of R

1. Install Rtools for Windows machines from CRAN Rtools.
2. Install the latest version of R from CRAN R.
3. Install the latest version of RStudio. RStudio is the editor we'll rely on, i.e. we'll write code in RStudio which is subsequently sent to and run within R.

Introduction to R

- What is R?
  - R is a powerful programming language and software environment designed for *data analysis*, *statistics*, *and visualization*. It is widely used by *scientists*, *researchers*, *and data analysts* to make sense of large amounts of data.
- Why Learn R?
  - Free and Open Source
  - Great for Data Analysis
  - Strong Visualization Tools
  - Widely Used in Industry

What can you do with R?

- Data management
  - Importing and exporting data in various formats
  - Cleaning and transforming data
  - Merging and appending data sets
- Data visualization
  - Scatter plot, histograms, PDFs and CDFs
  - Non-parametric plots
  - Regression discontinuity and dynamic DiD
- Data analysis
  - Linear regression
  - Panel data analysis
  - Time series analysis
  - Causal inference and structural analysis

Example: Working hours per worker

- The data set provides estimates of average annual working hours per worker across countries between 1950 and 2017 (Giattino, Ortiz-Ospina, and Roser, 2020)
- We will use this example to illustrate some R functions
  - Data management
    - Import an Excel file into R
    - Create a new variable for weekly working hours per worker
  - Data visualization
    - Plot a time-series graph
  - Data analysis
    - Use linear regression to examine whether workers in East Asia work longer

RStudio

R's Interface

1. *Source Pane* is a text editor that will be referred to as R Script
   - You can write commands and save them, which is the main point of reproducibility
2. *Console Pane* is the place you write your line-by-line command
   - It means you can only write a single command or a long paragraph of commands
   - After you close the RStudio, the commands will not be saved unless you specified to do so
   - Example: Symbol $>$ called prompt
     - Type $3 + 4$, and press Enter
3. *Environment Pane* has several sub-windows
   - Global Environment is the place where your data will be after importing data
   - History saves the commands you run in R console
   - Connections is where you connect to external databases

R's Interface

4. *Files Pane* also has several sub-windows
   - You can manage files and folders as well as set the working directory
   - Plots is where your plots will appear
   - Packages is where you manage your R packages - you can install it from CRAN or other repositories
   - Help is where R stores documentations - you can open the help or introduction page of the respective packages as well as individual functions
   - Example: ?lm() (in stock package 'stats')

R Scripts and R Markdown

1. R Scripts
   - Rather than typing everything into the console, this allows you to type and save your commands.
   - This can be manually edited with comments.
2. R Markdown
   - A versatile format for analysis and tracking progression.
   - Helpful for reproducibility.
   - Can be used to create various dynamic output files.

R Basics

Basics: packages and libraries

Packages only have to be loaded once

```
install.packages("tidyverse") # data transformation functions
```

Libraries have to be loaded each session

```
library(tidyr)
library(dplyr)
```

Basics: operators

Assignment

```
x <- 1
y <- 2
```

Arithmetic

```
x+y
x*y
```

Basics: operators

Comparison

```
x==y # equal
x!=y # not equal
x>y # greater than
x<=y # less than or equal to
```

Logical

```
y>x && y==2*x # AND: both conditions must be true
x > y || x < y # OR: either condition must be true
```

Basics: data types

Numeric

Double: decimal numbers

```
x <- 1.5
typeof(x) # returns object type
is.numeric(x) # returns if object is numeric or not
```

Integer: whole numbers

```
x <- 1L # L is necessary to make it an integer
```

Character

```
x <- "Hello"
```

```
x <- "5"
```

Perform an arithmetic operation on x now

Basics: data structure

Vectors: all elements must be of the same type

```
x <- sample(1:10, 5, replace = TRUE)
y <- sample(1:10, 5, replace = TRUE)
year <- 1990:1994
names <- c("payal", "amazon", "google", "microsoft", "alibaba")
```

Lists: can contain elements of different types and lengths

```
people <- list(name = c("bob", "jan", "mitsuha"),
               age = c(24, 36, 21),
               grade = c("1st", "2nd", "4th"))
```

Accessing elements in the list

```
people[1]
people[[1]]
people[[1]][1]
```

Basics: data structure

Exercise:

1. Create the following vectors
   - classes = classes you are taking
   - period = period of the corresponding classes
   - program = program you are in
   - year = year you are in
2. Store these vectors in a list
3. Pull from the list the period of this class (output should = 3)

Basics: data structure

Data Frames: two-dimensional of equal length (list of vectors)

```
df <- data.frame(names, year, x, y) # combines vectors into data frame
tibble(df) # see the data frame
```

Accessing the vectors (*variables*) in the data frame

```
df$year
```

```
df %>% pull(year) # piping: lecture 2
```

Homework 0

Homework 0: Set up your own Script/RMarkdown

1. Load the library readxl

```
install.packages("readxl")
library(readxl)
```

2. Import the data from the excel file and store it in the global environment

```
setwd("C://Users//...") # two forward slashes instead of back slash
data <- read_excel(file.path("annual-working-hours-per-worker.xlsx"),
                   sheet = "annual-working-hours-per-worker")
```

3. Skim the data with tibble: what are the data types of the variables?

```
tibble(data)
```

4. Render the file

```
rmarkdown::render("file_name.R", "pdf_document")
```